МИНИСТЕРСТВО ТРАНСПОРТА РОССИЙСКОЙ ФЕДЕРАЦИИ
ФЕДЕРАЛЬНОЕ АГЕНТСТВО ЖЕЛЕЗНОДОРОЖНОГО ТРАНСПОРТА
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**«ПЕТЕРБУРГСКИЙ ГОСУДАРСТВЕННЫЙ УНИВЕРСИТЕТ
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Кафедра «ИНФОРМАЦИОННЫЕ И ВЫЧИСЛИТЕЛЬНЫЕ СИСТЕМЫ»

Дисциплина: «Программирование С»

О Т Ч Е Т

по лабораторной работе № 2

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Цель Работы

Создать массив структур. Данными массива должны быть структуры соответствующего варианта. Создать функции для записи и чтения массива в текстовый и бинарный файлы . Реализовать минимум все функции, приведённые в лекции, предложить свои варианты функций работы со структурами. Создать меню для управления работы со структурами . В меню должны быть также пункты для сохранения данных на диске (имя файла запрашивается), чтения и просмотра файла, корректировки, добавления, работы с динамической памятью.

Постановка задачи 1

Листинг

main.c

```c
#include <stdio.h>
#include <stdlib.h>
#include "struct.h"
#include "funcs.h"


int main(){
  struct Club *clubs = NULL;
  int clbCnt = 0;

  int choise = 0;
  do {
  printf("\n");
  printf("\tChoose option\n1. Read File\n2. Show clubs\n3. Add club");
  printf("\n4. correct Club\n5. Write File\n6. Exit program\n");

  scanf("%d", &choise);
  switch (choise) {
    case 1:
      ReadFile(&clubs, &clbCnt);
      break;

    case 2:
      ShowClubs(clubs, clbCnt);
      break;

    case 3:
      AddClub(&clubs, &clbCnt);
      break;

    case 4:
      ChangeClub(clubs, clbCnt);
      break;

    case 5:
      WriteFile(clubs, clbCnt);
      break;

    case 6:
      printf("Exiting...\n");
      free(clubs);
      break;

    default:
      printf("WIP(or u enter wrong number)");
      break;
  }
  } while (choise != 6);
  return 0;
}
```

funcs.h

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include "struct.h"

void ReadFile(struct Club **arr, int *cnt);

void ShowClubs(struct Club *clbArr, int cnt);

void AddClub(struct Club **arr, int *count);

void ChangeClub(struct Club *arr, int cnt);

//-----------------------------------------------------------------

void WriteFile(struct Club *arr, int cnt);
```

funcs.c

```c
#include "funcs.h"
#include <stdio.h>

void ReadFile(struct Club **arr, int *cnt){
    char filename[128];
    printf("Please, Enter the filename\n");
    while (getchar() != '\n');
    fgets(filename, 128, stdin);

    filename[strcspn(filename, "\n")] = '\0';
    FILE *file = fopen(filename, "r");
    if (file == NULL){
        printf("file wasnt found");
        return;
    }

    int newClbsCnt = 0;
    fscanf(file, "%d\n", &newClbsCnt);

    if (*arr != NULL) {
      free(*arr);
  }

    struct Club *newArrClb = (struct Club*) malloc(newClbsCnt * sizeof(struct Club));

    for (int i = 0; i < newClbsCnt; i++){
        fscanf(file, "%[^;];", newArrClb[i].name);
        fscanf(file, "%[^;];", newArrClb[i].country);
        fscanf(file, "%d", &newArrClb[i].victoryCnt);

        for (int q = 0; q < newArrClb[i].victoryCnt; q++){
            fscanf(file, ";%d", &newArrClb[i].victoryYr[q]);
        }
        fscanf(file, "\n");
    }

    fclose(file);
    *arr = newArrClb;
    *cnt = newClbsCnt;
    return;
}

void ShowClubs(struct Club *clbArr, int cnt){
    if (cnt == 0){
        printf("list is empty");
        return;
    }


    printf("\n");
```

```
  printf("|  № | name                | Country             | Years
|\n");
  printf("|----|---------------------|---------------------|-------------------------
------|\n");
    for (int i = 0; i < cnt; i++) {
        printf("| %2d | %-19s | %-19s | ", i + 1, clbArr[i].name, clbArr[i].country);

        for (int j = 0; j < clbArr[i].victoryCnt; j++) {
            printf("%d", clbArr[i].victoryYr[j]);
          if (j < clbArr[i].victoryCnt - 1) {
            printf(", ");
          }
        }

        if (clbArr[i].victoryCnt == 0) {
            printf("Нет побед");
        }
        printf("\n");
    }

    return;
}
void AddClub(struct Club **arr, int *count) {
  struct Club new_club;

  while (getchar() != '\n');

  printf("Enter Club name: ");
  fgets(new_club.name, sizeof(new_club.name), stdin);
  new_club.name[strcspn(new_club.name, "\n")] = '\0';

  printf("Enter country: ");
  fgets(new_club.country, sizeof(new_club.country), stdin);
  new_club.country[strcspn(new_club.country, "\n")] = '\0';

  printf("Enter Count of Years of wins(up to 10): ");
  scanf("%d", &new_club.victoryCnt);

  if (new_club.victoryCnt > 10) {
      new_club.victoryCnt = 10;
      printf("UP TO 10\n");
  }

  if (new_club.victoryCnt > 0) {
      printf("Enter Years:\n");
      for (int i = 0; i < new_club.victoryCnt; i++) {
          printf("Year №%d: ", i + 1);
          scanf("%d", &new_club.victoryYr[i]);
      }
  }
```

```c
  // Перевыделение памяти
  struct Club *new_arr = (struct Club*)realloc(*arr, (*count + 1) * sizeof(struct
Club));
  if (new_arr == NULL) {
      printf("realloc error!\n");
      return;
  }

  new_arr[*count] = new_club;
  (*count)++;
  *arr = new_arr;
  printf("Club added succesful!\n");
}

//-----------------------------------------------------------------

void ChangeClub(struct Club *arr, int cnt){
    if (cnt == 0){
        printf("nothing to change");
        return;
    };

input_of_index:
    int clbToChange;
    printf("Enter number of club to change");
    scanf("%d", &clbToChange);

    if (clbToChange < 1 || clbToChange > cnt){
        printf("wrong club index");
        goto input_of_index;
    }

    clbToChange--;
    while (getchar() != '\n');

    printf("Editing the %s\n", arr[clbToChange].name);

    char buffer[100];

    printf("Pls, Enter new name (Return to save existing): ");
  fgets(buffer, sizeof(buffer), stdin);
  buffer[strcspn(buffer, "\n")] = '\0';
  if (strlen(buffer) > 0)
    strcpy(arr[clbToChange].name, buffer);

  printf("Pls, Enter new Country (Return to save existing): ");
  fgets(buffer, sizeof(buffer), stdin);
  buffer[strcspn(buffer, "\n")] = '\0';
  if (strlen(buffer) > 0)
      strcpy(arr[clbToChange].country, buffer);
```

```c
  char cnt_buf[20];
input_new_victory_cnt:
  printf("Pls, Enter new wins Count (Return to save existing): ");
  fgets(cnt_buf, sizeof(cnt_buf), stdin);
  cnt_buf[strcspn(cnt_buf, "\n")] = '\0';
  if (strlen(cnt_buf) > 0) {
      int new_cnt = atoi(cnt_buf);
      if (new_cnt <= 10) {
          arr[clbToChange].victoryCnt = new_cnt;
      } else {
          printf("UP TO 10\n");
          goto input_new_victory_cnt;
      }
  }

    if (arr[clbToChange].victoryCnt > 0) {
  printf("Pls, Enter new wins Count (Return to save existing): \n");
  for (int i = 0; i < arr[clbToChange].victoryCnt; i++) {
    printf("Victory Year №%d: ", i + 1);
    char year_buf[20];
    fgets(year_buf, sizeof(year_buf), stdin);
    year_buf[strcspn(year_buf, "\n")] = '\0';
      if (strlen(year_buf) > 0) {
        arr[clbToChange].victoryYr[i] = atoi(year_buf);
      }
      }
    }

  printf("Update successful!\n");
    return;
}

//-----------------------------------------------------------------

void WriteFile(struct Club *arr, int cnt){
  if (cnt == 0) {
    printf("\nNothing to save.\n");
    return;
  }

  char filename[128];
  printf("Pls, Enter filename: ");

  while (getchar() != '\n');
  fgets(filename, 128, stdin);
  filename[strcspn(filename, "\n")] = '\0';

  FILE *file = fopen(filename, "w");
  if (file == NULL) {
    printf("Error open %s to write!\n", filename);
    return;
```

```c
  }

  fprintf(file, "%d\n", cnt);

  for (int i = 0; i < cnt; i++) {
    fprintf(file, "%s;%s;%d",
      arr[i].name,
      arr[i].country,
      arr[i].victoryCnt);

      for (int j = 0; j < arr[i].victoryCnt; j++) {
        fprintf(file, ";%d", arr[i].victoryYr[j]);
      }
      fprintf(file, "\n");
  }

  fclose(file);
  printf("Save successful to file %s\n", filename);

    return;
}
```

struct.h

```c
#pragma once
struct Club{
  char name[20];
  char country[20];
  int  victoryYr[10];
  int  victoryCnt;
};
```

компиляция

```
gcc -c src/main.c -o src/obj/main.o
gcc -c src/funcs.c -o src/obj/funcs.o
gcc src/obj/funcs.o src/obj/main.o -o bin/progr
```

ВЫВОД

```
→  bin git:(T2c2) ✗ ./progr

        Choose option
1. Read File
2. Show clubs
3. Add club
4. correct Club
5. Write File
6. Exit program
2
list is empty
        Choose option
1. Read File
2. Show clubs
3. Add club
4. correct Club
5. Write File
6. Exit program
3
Enter Club name: PSG
Enter country: France
Enter Count of Years of wins(up to 10): 2
Enter Years:
Year №1: 2027
Year №2: 2028
Club added succesful!

        Choose option
1. Read File
2. Show clubs
3. Add club
4. correct Club
5. Write File
6. Exit program
2

|  № | name                | Country             | Years                           |
|----|---------------------|---------------------|---------------------------------|
|  1 | PSG                 | France              | 2027, 2028

        Choose option
1. Read File
2. Show clubs
3. Add club
4. correct Club
5. Write File
6. Exit program
```

```
→  bin git:(T2c2) ✗ ./progr

        Choose option
1. Read File
2. Show clubs
3. Add club
4. correct Club
5. Write File
6. Exit program
3
Enter Club name: PSG
```

```
Enter country: France
Enter Count of Years of wins(up to 10): 2
Enter Years:
Year №1: 2027
Year №2: 2028
Club added succesful!

        Choose option
1. Read File
2. Show clubs
3. Add club
4. correct Club
5. Write File
6. Exit program
5
Pls, Enter filename: qwe
Save successful to file qwe

        Choose option
1. Read File
2. Show clubs
3. Add club
4. correct Club
5. Write File
6. Exit program
6
Exiting...
→  bin git:(T2c2) ✗ ./progr

        Choose option
1. Read File
2. Show clubs
3. Add club
4. correct Club
5. Write File
6. Exit program
1
Please, Enter the filename
qwe

        Choose option
1. Read File
2. Show clubs
3. Add club
4. correct Club
5. Write File
6. Exit program
2

|  №  | name                | Country             | Years                          |
|----|---------------------|---------------------|--------------------------------|
|  1  | PSG                 | France              | 2027, 2028

        Choose option
1. Read File
2. Show clubs
3. Add club
4. correct Club
5. Write File
```